Use Case Full Description:

Use Case Name:	WAH4PCE INTEROPERABILITY
Scenario:	A health data Provider submits patient records or clinical data into the WAH4PCE Interoperability platform. Depending on the native format of the incoming data, an AI agent may need to translate it into standardized formats (HL7 FHIR R4/HL7 v2), which subsequently triggers a mandatory privacy consent validation.
Triggering Event:	The Provider initiates a data transfer, upload sequence, or API call from their local system to the WAH4PCE interoperability endpoint.
Brief Description:	This use case defines the process by which a Provider supplies health data to the ecosystem. It manages core data ingestion while seamlessly handling conditional data translation and strict privacy consent verification, ensuring that shared data is both interoperable and legally compliant.
Actors:	**Primary:** Provider **Secondary:** WAH4PCE AI
Related Use Cases:	• Translate Data to/from PH Core HL7 FHIR R4/HL7 v2(<<Extend>>) • **Validate Privacy Consent** (<<Include>> via the Translate Data use case)
Stakeholders:	• **Provider:** Requires a secure, reliable, and frictionless data transmission pathway. • **Requestor:** Depends on the availability, standardization, and accuracy of the provided data. • **WAH4PCE Admin:** Monitors system uptime, API health, and data flow integrity. • **Patient (Implicit):** Requires their health data to be handled securely and strictly according to their privacy consent preferences.
Preconditions:	• Provider must be authenticated and authorized to access the WAH4PCE integration endpoint. • The patient whose data is being submitted must exist or be identifiable within the system.

<table>
<tr><td></td><td colspan="2">The WAH4PCE interoperability platform and AI translation services must be online and operational.</td></tr>
<tr><td>Postconditions:</td><td colspan="2">Health data payload must be standardized to PH Core HL7 FHIR R4 or HL7 v2 format.
Patient privacy consent must be verified and an audit trail of the validation must be logged.
The standardized and consent-validated data must be securely committed to the interoperable repository.
A success status notification must be generated and transmitted back to the Provider.</td></tr>
<tr><td rowspan="2">Flow of Activities:</td><td>Actor</td><td>System</td></tr>
<tr><td>1. Provider initiates the transaction by submitting a health data payload to the WAH4PCE integration endpoint.

4. WAH4PCE AI intercepts the workflow and executes the Translate Data to/from PH Core HL7 FHIR R4/HL7 v2 use case.</td><td>2. System receives the data payload, encrypts it in transit, and performs initial structural and security validation.

3. System analyzes the incoming data format. If the data is non-standard and requires normalization to PH Core HL7 FHIR R4 or HL7 v2, the system hits an extension point.

5. As a mandatory inclusion of the translation and processing pipeline, the system triggers the Validate Privacy Consent use case to verify the patient's current data-sharing permissions.

6. System compiles the translated data and the consent validation results. If consent is verified as active and valid, the system commits the standardized data to the interoperable repository.</td></tr>
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		7. System generates and transmits a success status notification back to the Provider.
Exception Conditions:		
E1: Corrupted or Malformed Payload	If the system detects a structurally invalid payload at Step 2, it immediately aborts the transfer, logs the event for administrative review, and returns a descriptive 400 Bad Request or equivalent error code to the Provider.	
E2: AI Translation Mapping Failure	If the WAH4PCE AI encounters anomalous data that it cannot accurately map to HL7 FHIR R4/HL7 v2 during Step 4, the transaction is safely halted. The data is temporarily quarantined, and an alert is dispatched to both the Provider and the WAH4PCE Admin.	
E3: Privacy Consent Denied or Revoked	If the *Validate Privacy Consent* check (Step 5) fails, indicates an opt-out, or cannot find matching consent records, the system instantly purges the payload from active memory, prevents repository storage, logs an immutable audit trail of the rejection, and notifies the Provider of a consent-based denial.	